Scheme/ Answer Key for Valuation

Scheme of evaluation (marks in brackets) and answers of problems/key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.TECH S2 (R,S)EXAMINATION(2019 SCHEME), JUNE 2022

Course Code: PHT 100

Course Name: ENGINEERING PHYSICS A

(2019 -Scheme)

Max. Marks: 100 | Duration: 3 Hours

PART A

		Answer all questions, each carries 3 marks	Marks
1		Equation connecting relaxation time and damping factor &explanation of relaxation time inversely proportional to damping –1 ½ Marks; Equation relating Quality factor and relaxation time and explanation of Quality factor directly proportional to relaxation time and inversely proportional to damping…. 1 ½ Marks.	(3)
2		Three laws- 3 Marks	(3)
3		Condition for minima and maxima ---0.5 Mark; Explanation ---2 Marks, example – 0.5 Mark.	(3)
4		Definition -1 Mark, comparison with any two points -2 Marks	(3)
5		Equation of energy $E_n = n^2h^2/8mL^2$or $n^2\pi^2\hbar^2 /2mL^2$ with correct notations ---2 Marks , n = 2 and equation for first excited energy state-1Mark	(3)
6		Quantum dot -1½ Marks, quantum wire -1½ Marks	(3)
7		Ampere's circuital theorem - 2 Marks, mathematical statement – 1Mark	(3)
8		Divergence -1½ Marks, physical significance-1 ½ Marks	(3)
9		Definition – 2 Marks, Two examples – 1 Mark	(3)
10		Figure- 1 Mark, working-2 Marks	(3)

PART B

Answer one full question from each module, each question carries 14 marks.

MODULE 1

11		a) Writing the differential equation and solution- 2 Marks; Conditions and explanation of three cases – 6.5 Marks (Overdamped – 2 Marks, Critically damped – 2 Marks, Underdamped – 2.5 Marks); Displacement – Time graph of three cases--- 1.5 Marks, b) writing the differential equations of an LCR circuit and mechanical oscillator - 1 Mark.	(10) (4)

		Comparison of electrical oscillator with mechanical oscillator (any 3 points) - 3 Marks	
12		a) figure -1 Mark, explanation --- 2 Marks, derivation of one dimensional wave equation- 4 Marks, general equation of one dimensional wave-1 Mark, by comparison obtain the equation v = $\sqrt{\frac{T}{m}}$ - 2 Marks b) Amplitude-1 Mark, Time period-1 Mark, velocity – 1 Mark wavelength – 1 Mark A=0.25×10^{-3} m, λ=251.2 m, T=0.0126 sec, v = 2×10^{4}m/s	(10) (4)
MODULE 2			
13		a) Figure – 1 Mark, Construction and band formation-3 Marks, Derivation of radius of n th ring - 4 Marks, Obtaining the expression for wavelength - 2 Marks b) Antireflection coating-1 mark, Minimum thickness t = $\lambda/4\mu_f$ -1 Mark, calculation and result with unit -2 Marks t = 0.102μm	(10) (4)
14		a) Resolving power- 2 Marks, Figure – 1 Mark , Diffraction pattern formation with grating -2 Marks, Derivation of Grating equation – 5 Marks,diagram-1 Mark b) Equation Sinθ =nNλ -1 Mark, First order θ_1 = 19.27-1 Mark Second order θ_2 = 41.30˚-1 Mark, Angular separation =22.03˚-1 Mark	(10) (4)
MODULE 3			
15		a) Plane wave equation-1 mark, $P^2\psi$- 1 Marks, Eψ -1 Marks, time dependent equation-1Mark wavefunction ψ is a product of two functions-1 Mark , time independent equation- 5 Marks b) Heisenberg's uncertainty principle and its equations-2 Marks Natural line broadening –explanation-2 Marks	(10) (4)
16		a) Nanomaterials -2 Marks Mechanical, optical, electrical properties of nanomaterials-6 Marks (2 Marks each) b) Six applications-6 Marks	(8) (6)
MODULE 4			
17		a) Magnetization, Magnetic permeability, Relative permeability, Susceptibility – 8 Marks (2 Marks each), Derivation of relation between χ and μ_r – 2 Marks b) Induced emf at 5sec = $d\phi_B/dt$ = 52V- 2 Marks (Equation – 1 Mark, Final Answer with unit – 1 Mark)	(10) (4)

		The induced current in 5 sec = V/R =52/2.5 =20.8 A - 2 Marks.(Equation – 1 Mark, Final Answer with unit -1 Mark)	
18		a) Maxwell's equations in free space -2 marks derivation of one dimensional wave equation using Maxewell's equations-4marks Compare it with general equation and write velocity equation -2marks Velocity equal to velocity of light – 2 marks b) Explanation of Poynting's theorem and write Poynting's vector-4 marks	(10) (4)
MODULE 5			
19		a) Definition critical temperature ,critical field – 2 Marks each (4 Marks) Relation between them – 1 Mark BCS theory – 5 Marks b) Explanation of Meissner effect with figure – 2 Marks Derivation of superconductors are diamagnetic (χ = -1) – 2 Marks	(10) (4)
20		a) Block diagram – 3 Marks, Explanation of parts – 3 Marks Four advantages of Fibre optic communication- 4 Marks b) Equations NA = $\sqrt{n1^2 - n2^2}$-1 Mark n_1-n_2 = 0.05, NA = 0.38 -1 mark , substitution and calculation -1 Mark refractive index of the core n_1 = 1.469 -1 Mark	(10) (4)
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